

| Traceability Matrix |                                                                                                                                                                                                                                                                                                                    |             |                                                                                                                                                                                                                                                                                                                         |             |                                                                                                                                                                                                                                                                                    |                                 |            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|------------|
| FeatureId           | Feature                                                                                                                                                                                                                                                                                                            | SystemReqId | System Requirement                                                                                                                                                                                                                                                                                                      | UserStoryId | User Story                                                                                                                                                                                                                                                                         | Major System Component          | Completion |
| F1                  | The system will allow users and restaurant owners/managers to create accounts and log in. It will handle user authentication (login), authorization (what each user is allowed to do), and basic user identity management. Different roles such as regular users and restaurant owners/managers will be supported. | SR1-1       | The system shall allow users to create an account using a username, email and password.                                                                                                                                                                                                                                 | US1-1       | As a user, I want to create an account in the iFody app using my username, email, and password, so that I am able to access the service being provided                                                                                                                             | User management component       | SR1-1      |
|                     |                                                                                                                                                                                                                                                                                                                    | SR1-2       | The system shall allow users (customers and owners) to log into their account with the username or email and password if they have already created an account.                                                                                                                                                          | US1-2       | As a user, I want to log into my account using my username or email and password, so that I do not need to create a new account every time to access the service.                                                                                                                  | User Management Component       | SR1-2      |
|                     |                                                                                                                                                                                                                                                                                                                    | SR1-3       | The system shall keep track of user roles and format their experience accordingly with the proper tools at their disposal.                                                                                                                                                                                              | US1-3       | As a customer, I want my interface to allow me to order food so that I can use the app according to my needs.                                                                                                                                                                      | User management component       | SR1-3      |
|                     |                                                                                                                                                                                                                                                                                                                    |             |                                                                                                                                                                                                                                                                                                                         | US1-4       | As a manager/owner, I want to be able to edit my menus for my restaurants and set different prices so that I can keep the information accurate based on my manager role                                                                                                            | User management component       |            |
| F2                  | The system will store information about restaurants and their menus. It will ensure that data is valid, properly connected (for example, menu items must belong to a restaurant), and that basic constraints are enforced, such as preventing invalid or missing values.                                           | SR2-1       | The system shall remain up to date with menu items through period-based refreshings.                                                                                                                                                                                                                                    | US2-1       | As a manager/owner, I want to keep my menu items up to date so I can advertise our new foodmeal items.                                                                                                                                                                             | Restaurant Management Component | SR2-1      |
|                     |                                                                                                                                                                                                                                                                                                                    |             |                                                                                                                                                                                                                                                                                                                         | US2-2       | As a customer, I want menus to be up to date so that I don't order something that doesn't exist and/or miss something that is not shown.                                                                                                                                           | Restaurant Management Component |            |
|                     |                                                                                                                                                                                                                                                                                                                    | SR2-2       | The system shall store information mapped to the correct restaurants.                                                                                                                                                                                                                                                   | US2-3       | As a manager/owner, I want to see my restaurant's information page on the app correctly show my real-time restaurant status so the customers will not have misguided impressions of my restaurant.                                                                                 | Restaurant Management Component | SR2-2      |
|                     |                                                                                                                                                                                                                                                                                                                    |             |                                                                                                                                                                                                                                                                                                                         | US2-4       | As a customer, I want the online statuses of the restaurants to match correctly with the real offline statuses, so I can trust the application more.                                                                                                                               | Restaurant Management Component |            |
| F3                  | The system will allow users to browse restaurant menus and search for items or restaurants. Backend logic will handle filtering, searching, and returning paginated results.                                                                                                                                       | SR3-1       | The system shall display a list of restaurants for the user to choose (perhaps the list is based on some sort of algorithm that uses user purchase history to recommend restaurants or items)                                                                                                                           | US3-1       | As a customer, I would like to know what restaurants are offered on the app so I can make a fully informed decision on where I would like to order from.                                                                                                                           | Browse Restaurant Component     | SR3-1      |
|                     |                                                                                                                                                                                                                                                                                                                    | SR3-2       | The system shall integrate a searching function for the user to find the restaurants or items they want directly                                                                                                                                                                                                        | US3-2       | As a customer, I want to be able to search for the restaurants I would like to order from and to browse their menu for the food I would like.                                                                                                                                      | Browse Restaurant Component     | SR3-2      |
|                     |                                                                                                                                                                                                                                                                                                                    | SR3-3       | The system shall filter any restaurants that are out of the user's delivery range                                                                                                                                                                                                                                       | US3-3       | As a customer, I want to only be able to order from places within my reach, so no order cancellation happens once my order is completed.                                                                                                                                           | Browse Restaurant Component     | SR3-3      |
| F4                  | The system will allow users to create and manage food orders. It will ensure that orders are consistent, correctly stored, and follow business logic for the domain (for example, an order cannot be modified after it is completed).                                                                              | SR4-1       | The system shall allow the customers to create food orders among the available restaurants and items.                                                                                                                                                                                                                   | US4-1       | As a customer I want to create an order under my selected restaurant so I will not be limited solely on browsing the restaurant description page.                                                                                                                                  | Order Management Component      | SR4-1      |
|                     |                                                                                                                                                                                                                                                                                                                    | SR4-2       | The system shall allow the user to cancel or modify the orders without any punishment if the order was not confirmed yet on the restaurant side.                                                                                                                                                                        | US4-2       | As a customer, I want to be able to fix a miss-click on an order or be able to modify an item or order if I quickly change my mind.                                                                                                                                                | Order Management Component      | SR4-2      |
|                     |                                                                                                                                                                                                                                                                                                                    | SR4-3       | The system should not allow for an order to be modified after it has been completed.                                                                                                                                                                                                                                    | US4-3       | As a manager/owner, I want to be certain that once the order is sent to the customer, it can not be modified anymore, so no waste happens.                                                                                                                                         | Order Management Component      | SR4-3      |
| F5                  | The system will manage delivery-related information. It will support assigning deliveries and tracking basic delivery status as part of the backend logic.                                                                                                                                                         | SR5-1       | The system shall notify the user when the food has left the restaurant, when it arrives at its destination, and gives an accurate estimation of the time until delivery when the order is processed.                                                                                                                    | US5-1       | As a customer I want to know how long it will take for my order to get to me so that I know how to plan my time, and when I have to meet the driver.                                                                                                                               | Track Delivery Component        | SR5-1      |
|                     |                                                                                                                                                                                                                                                                                                                    |             |                                                                                                                                                                                                                                                                                                                         | US5-2       | As a manager, I want my employees to be held to a timeline, to ensure good customer service and overall satisfaction.                                                                                                                                                              | Track Delivery Component        |            |
| F6                  | The system will calculate the total cost of an order, including item prices, delivery fees, and taxes. These calculations will follow predefined business rules implemented in the backend.                                                                                                                        | SR6-1       | The system shall be able to calculate the sum of the prices of all the items being added to cart, and add the additional necessary taxes based on the order total.                                                                                                                                                      | US6-1       | As a user, I want to see the final price of the items I have ordered with tax so I know how much I need to pay via the platform.                                                                                                                                                   | Order Management Component      | SR6-1      |
|                     |                                                                                                                                                                                                                                                                                                                    | SR6-2       | The system shall be able to distinguish each restaurant's delivery fees and place the correct delivery fee on each order.                                                                                                                                                                                               | US6-2       | As a manager, I want to see the final price of the order with each single item's price correctly appearing, so I can check the prices with the offline menu to see all the figures are matched.                                                                                    | Order Management Component      | SR6-2      |
| F7                  | The system will simulate payment processing. No real payment gateway will be used, but the system will follow the correct workflow for accepting or rejecting a payment and updating the order status.                                                                                                             | SR7-1       | The system shall have multiple payment methods for the customers to choose, like via credit card, debit card, cash, etc.                                                                                                                                                                                                | US7-1       | As a customer I want to switch to a different payment method if I cannot provide the given payment method, so when I only have cash, I can switch from "pay by credit card" to "pay by cash" if the default method is "pay by credit card".                                        | Payment Processing Component    | SR7-1      |
|                     |                                                                                                                                                                                                                                                                                                                    | SR7-2       | The system shall have a cancel button for users to cancel the order if they cannot provide a valid payment method.                                                                                                                                                                                                      | US7-2       | As a customer I want to be able to cancel my order after a certain amount of time, so that in case I made a mistake in my order or in case I decide I do not want to order anymore, I can prevent the order process from completing.                                               | Payment Processing Component    | SR7-2      |
|                     |                                                                                                                                                                                                                                                                                                                    | SR7-3       | The system shall update the order status to active as a need-to-complete task for the restaurant if a payment is accepted.                                                                                                                                                                                              | US7-3       | As a manager, I want to be able to see newly assigned orders so I can notify the chefs to start preparing for the meals.                                                                                                                                                           | Payment Processing Component    | SR7-3      |
|                     |                                                                                                                                                                                                                                                                                                                    |             |                                                                                                                                                                                                                                                                                                                         | US7-4       | As a customer, I want to see the order is successfully placed at the restaurant if my payment is successful so I know eventually the meal will be delivered.                                                                                                                       | Payment Processing Component    |            |
| F8                  | Feature 8: The system will generate notifications or events when important actions occur, such as order creation or status changes.                                                                                                                                                                                | SR8-1       | The system shall provide the user with continuous updates of their order, such as when the order is received, the order is being prepared, the delivery driver is on their way to the restaurant to pick up the order, the delivery driver has picked the order and is on their way, and when the order gets delivered. | US8-1       | As a customer, I want to receive continuous feedback on the status of my order, so that I am aware of what phase in the order's process my order is in.                                                                                                                            | Track Delivery Component        | SR8-1      |
|                     |                                                                                                                                                                                                                                                                                                                    |             |                                                                                                                                                                                                                                                                                                                         | US8-2       | As a manager, I want to receive continuous information on the orders my restaurant is receiving, so that I know when to start preparing the food, when the driver is on their way to pick up the food and if the order was successfully completed by the delivery driver.          | Track Delivery Component        |            |
|                     |                                                                                                                                                                                                                                                                                                                    | SR8-2       | The system shall generate notifications in the event of changes in the order such as additional items, special requests or change in delivery address.                                                                                                                                                                  | US8-3       | As a customer, I want to receive notifications if/when I decide to make changes to my order, updating me if the changes I've made were successful and are also being processed, so that I can confirm that the changes/updates I've made have been acknowledged by the restaurant. | Track Delivery Component        | SR8-2      |
|                     |                                                                                                                                                                                                                                                                                                                    |             |                                                                                                                                                                                                                                                                                                                         | US8-4       | As a manager, I want to receive notifications if/when a customer decides to make changes to their orders, so that my restaurant can make the necessary adjustments to the order in a timely and organized manner.                                                                  | Track Delivery Component        |            |
|                     | Feature 9 (Optional): The system may allow users to rate and review                                                                                                                                                                                                                                                |             | The system shall allow for users to                                                                                                                                                                                                                                                                                     | US9-1       | As a user, I would like to be able to provide a review and a rating of my order so that I can provide the restaurant with feedback and so that other users can make informed decisions based on previous customer feedback.                                                        | Review Management Component     |            |

|                                               |                                                                                                                                                                                                                                          |              |                                                                                                                                                                                                        |                        |                                                                                                                                                                                                                                                                                                                                                                                                         |                              |              |  |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|--------------|--|
| F9                                            | completed orders or restaurants. This feature is optional and can be implemented if time permits.                                                                                                                                        | SR9-1        | write reviews and provide a rating of their order once the order is completed.                                                                                                                         | US9-2                  | As a manager, I would like to receive ratings and reviews from customers that have ordered from my restaurant, so that I can improve the lacking areas of my restaurant and so that previous users' ratings can serve as complementary marketing for my restaurant.                                                                                                                                     | Review Management Component  | SR9-1        |  |
| F10                                           | Feature 10 (Optional): The system may include administrative features such as viewing all orders and generating simple reports or statistics (for example, average delivery time or most popular restaurants). This feature is optional. | SR10-1       | The system shall continuously update and portray a report spreadsheet of administrative features of all the orders that have occurred in the app.                                                      | US10-1                 | As a developer of the iFody app, I would like to be able to generate a weekly food delivery time consumption report via clicking a 'generate report' button, so I can assign more delivery assistants to popular restaurants, and gain valuable information about site performance and consistency between orders.                                                                                      | Admin Management Component   | SR10-1       |  |
| M4 Features                                   |                                                                                                                                                                                                                                          |              |                                                                                                                                                                                                        |                        |                                                                                                                                                                                                                                                                                                                                                                                                         |                              |              |  |
| F-Trey                                        | Personal Feature (Trey): The system should display on the homepage, algorithm based restaurant lists based on user preferences in the area and recent orders                                                                             | SR-TREY-1    | The system shall display personal recommendations to the user based on an algorithm for popular restaurants in the current location and based on the active user's recent orders.                      | US-TREY-1<br>US-TREY-2 | As a customer I would like to get personal recommendations for popular restaurants in the area I am in, so that no matter my knowledge of restaurants in the area I will always get good recommendations,<br><br>As a customer, I would like a fast way to reorder from a restaurant I order from frequently to save time searching for it and to be able to pick from restaurants that I know I enjoy. | Browse Restaurant Component  | SR-TREY-1    |  |
| F-Aaron                                       | Personal Feature (Aaron):                                                                                                                                                                                                                | SR-AARON-1   | The system shall displays offers that refresh weekly, which only one of them can be activated at the same time, with pricing adjustments applying on specific items once a matched offer is activated. | US-Aaron-1             | As a customer, I would like to activate an offer that applies to my cart items, so I can buy the items with a lower price.                                                                                                                                                                                                                                                                              | Offer Service Component      | SR-AARON-1   |  |
| F-Umberto                                     | Personal Feature (Umberto): The system now has a new payment option that uses fake currency to complete transactions (PayPal Sandbox)                                                                                                    | SR-UMBERTO-1 | The system shall allow an external payment option - payment processing through PayPal -, allowing users to do secure payment transactions to a real payment provider.                                  | US-UMBERTO-1           | As a customer, I would like to be able to perform secure payment transactions with a real payment provider                                                                                                                                                                                                                                                                                              | Payment Processing Component | SR-UMBERTO-1 |  |
| F-Joao                                        | Personal Feature (Joao): The system now sends an email to the user whenever an order has been completed under the user's email                                                                                                           | SR-JOAO-1    | The system shall send the customer an email whenever their order is delivered, including the price, items, and a message thanking the customer for choosing iFody                                      | US-JOAO-1              | As a customer, I would like to receive an email when my order has been completed, so that I have a receipt outside the app and another place to store information about my order and delivery                                                                                                                                                                                                           | Track Delivery Component     | SR-JOAO-1    |  |
| F-Josh                                        | Personal Feature (Josh): The system now offer an AI chatbot to assist user to find available restaurant items based on what the customer wants to eat.                                                                                   | SR-JOSH-1    | The system shall provide a AI chatbot that answer customer's question about what restaurants are available and recommended items based on customer's prompt.                                           | US-JOSH-1              | As a customer, I want to receive quick answers about available restaurants and recommended food dishes without going through the whole restaurant pages.                                                                                                                                                                                                                                                | AI Assistant Component       | SR-JOSH-1    |  |
| Extensive Features outside of the scope of M3 |                                                                                                                                                                                                                                          |              |                                                                                                                                                                                                        |                        |                                                                                                                                                                                                                                                                                                                                                                                                         |                              |              |  |
| F10-EXTRA                                     | F10 does not specify anything to do with user management from an admin perspective                                                                                                                                                       | SR10-2       | The system shall allow the admin to block, unblock, and delete users, allowing for better system control                                                                                               | US10-2                 | As an admin, I would like to be able to block, unblock, and delete users so that I can maintain system integrity and handle misuse appropriately.                                                                                                                                                                                                                                                       | Admin Management Component   | SR10-2       |  |
| F4-EXTRA                                      | F4 has no requirement for users viewing their past orders                                                                                                                                                                                | SR4-4        | The system shall allow the user to be able to see their past orders, allowing the user to know the orders delivered to them and which orders the transaction failed                                    | US4-4                  | As a customer, I would like to keep track of the orders delivered to me and those orders for which the payment failed to go through, along with associated order items, and timestamps                                                                                                                                                                                                                  | Order Management Component   | SR4-4        |  |
| F6-EXTRA                                      | F6 only specifies basic order calculation for cart items. It does not contain any extensive requirements or directions for showing changes in price with situations such as adding an offer to an order.                                 | SR6-3        | The system shall check if the applied items of an active offer are in the cart and the order, and apply the price reduction according to the type of the offer.                                        | US6-3                  | As a customer, I would like to check the total price before and after offer activation so I know whether or not the offer has been applied to the items                                                                                                                                                                                                                                                 | Offer Service Component      | SR6-3        |  |
| F7-EXTRA                                      | F7 has no requirements for viewing past transactions.                                                                                                                                                                                    | SR7-4        | The system shall allow the user to see its past failed/completed transactions, allowing them to track their payment history and be certain about the amount they were charged while using the system   | US7-5                  | As a customer, I would like to be able to keep track of my past transactions to know how much I was charged for each completed order                                                                                                                                                                                                                                                                    | Payment Processing Component | SR7-4        |  |
| F8-EXTRA                                      | F8 has no specification for tracking read and unread notifications as well as a UI to track these status' in app                                                                                                                         | SR8-3        | The system shall allow the user to track their notification status - read and unread - within the app interface, allowing for better control and a more organized notification system                  | US8-5                  | As a customer, I want to see which notifications are read and unread so that I can easily identify new updates                                                                                                                                                                                                                                                                                          | Track Delivery Component     | SR8-3        |  |
| F1-EXTRA                                      | F1 has no specification on separate pages for different roles                                                                                                                                                                            | SR1-4        | The system shall dynamically adjust the frontend interface based on the authenticated user's role, displaying only the features and views relevant to that role                                        | US1-5                  | As a User on the platform I would like a frontend view tailored to my purposes on the sight. I do not want to see admin features greyed out as a customer. I want a simple and clean interface with no distractions so that I can order efficiently and get my food the fastest I can                                                                                                                   | User management component    | SR1-4        |  |